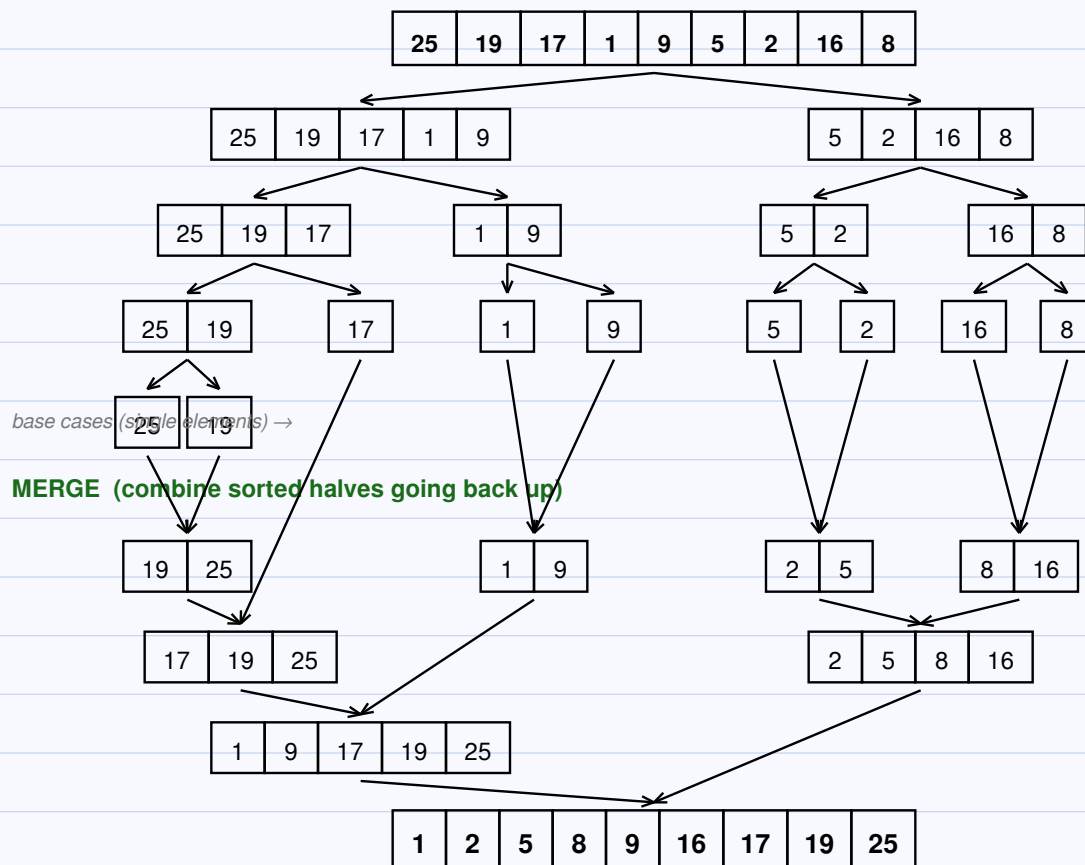


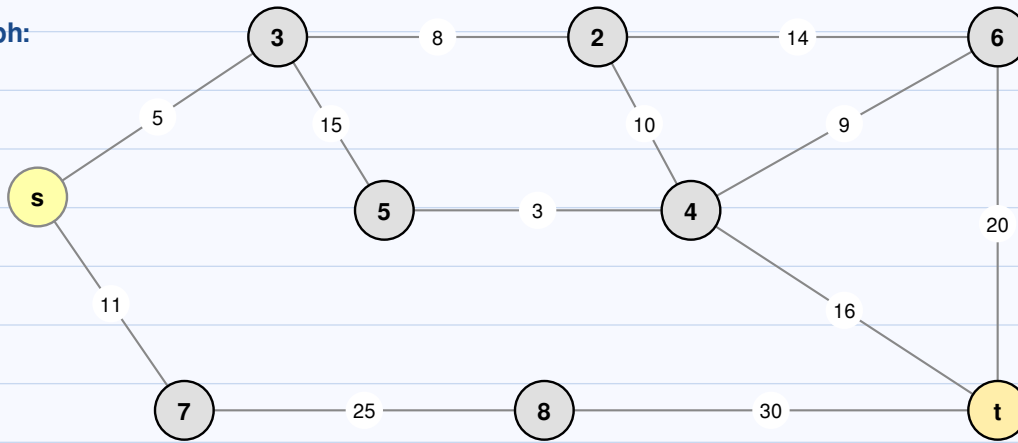
(20 pts) 1. MergeSort recursion tree on: [25, 19, 17, 1, 9, 5, 2, 16, 8]

Show divide phase (splitting) and merge phase (combining sorted halves).

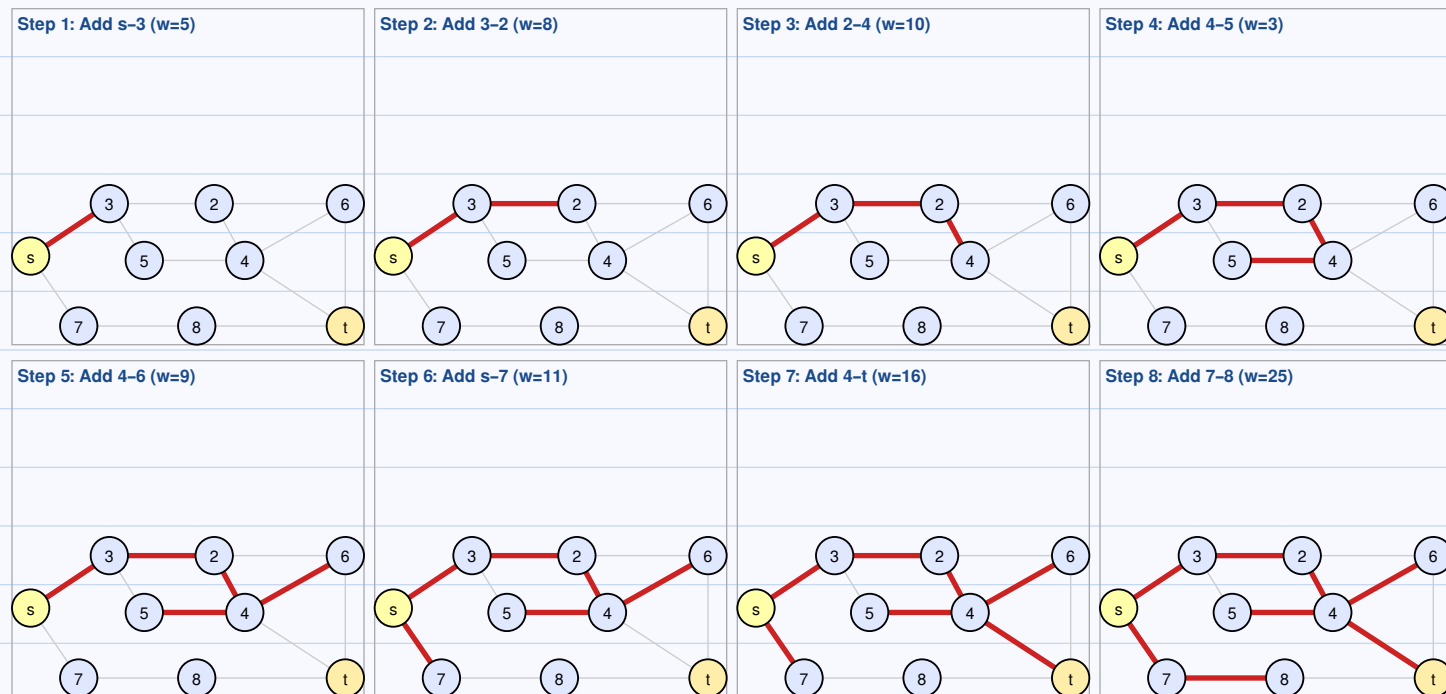
DIVIDE (split in half each level)

(60 pts) 2. Find the MST using Prim's, Kruskal's and Reverse Delete algorithms.

Graph:



Prim's Algorithm — step-by-step (start from s):



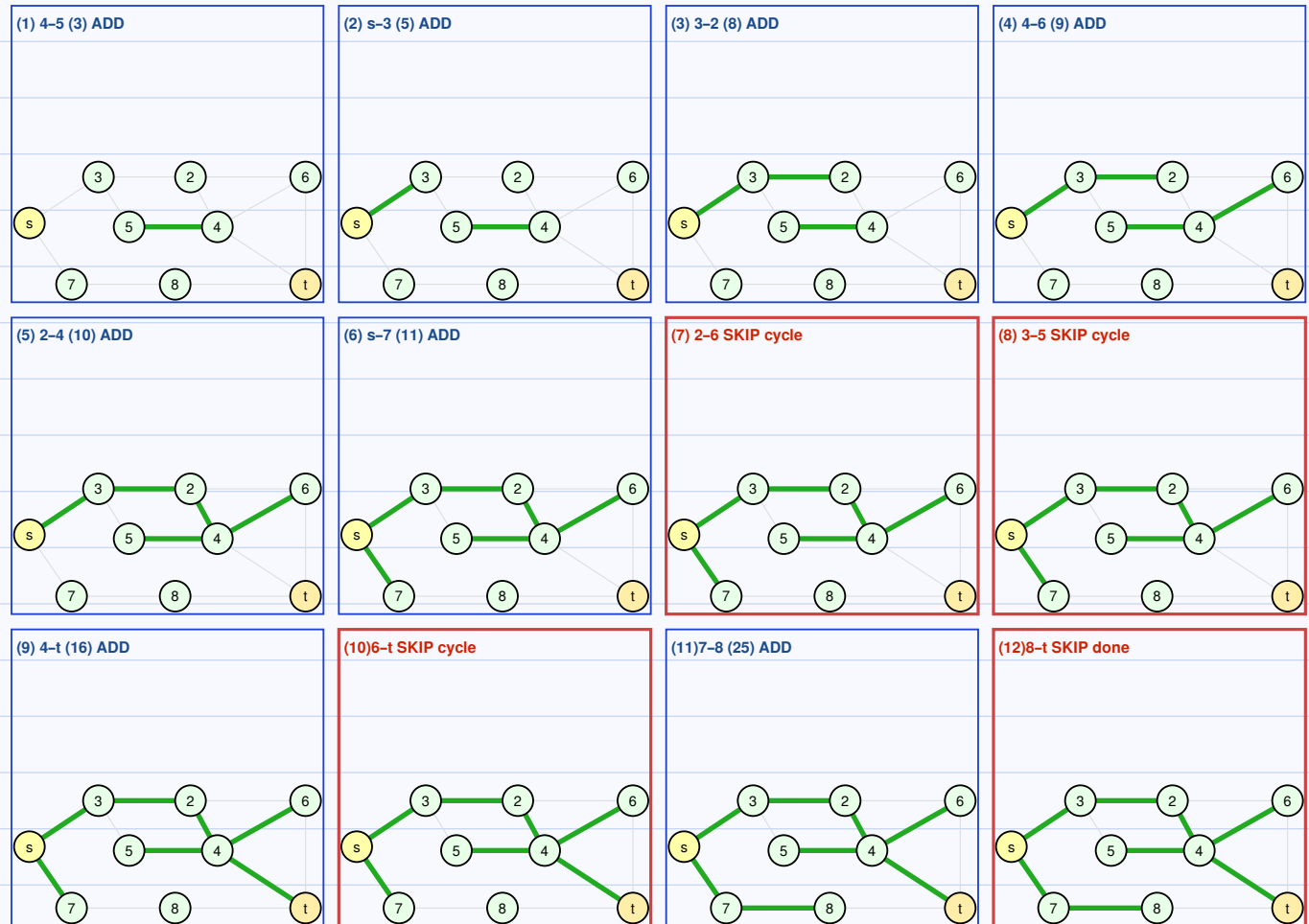
Prim's — Edge Addition Table:

Step	Edge	Wt	Vertex added	Why chosen
1	s-3	5	s ← 3	min edge from s
2	3-2	8	add 2	next min crossing
3	2-4	10	add 4	$2-4 < 3-5(15)$
4	4-5	3	add 5	smallest crossing!
5	4-6	9	add 6	$4-6 < 2-6(14)$
6	s-7	11	add 7	s-7 only option
7	4-t	16	add t	$4-t < 6-t(20)$
8	7-8	25	add 8	last vertex; done!

Prim's MST Total Weight = $5+8+10+3+9+11+16+25 = 87$

Kruskal's Algorithm — sort all edges by weight, add if no cycle (union-find)

Sorted edges: 4-5(3) s-3(5) 3-2(8) 4-6(9) 2-4(10) s-7(11) 2-6(14) 3-5(15) 4-t(16) 6-t(20) 7-8(25) 8-t(30)

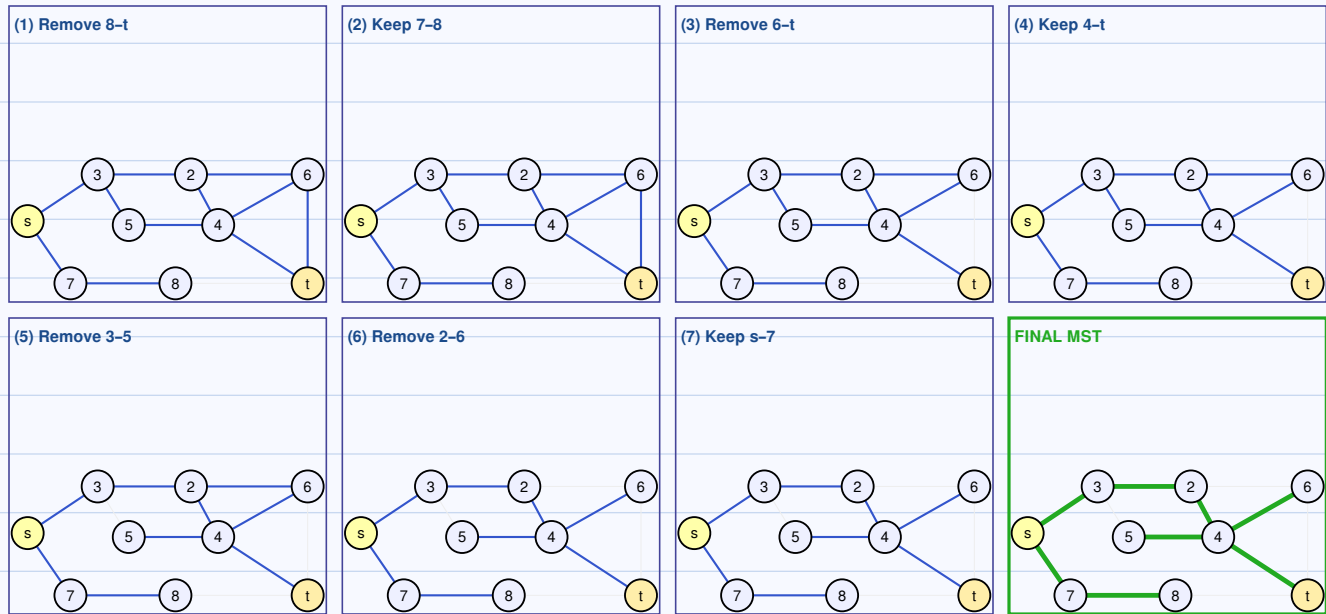
**Kruskal's — Edge Decision Table:**

Stp	Edge	Wt	Cycle?	Action	Components
1	4-5	3	No	ADD	{s}{3}{2}{6}{4,5}{7}{8}{t}
2	s-3	5	No	ADD	{s,3}{2}{6}{4,5}{7}{8}{t}
3	3-2	8	No	ADD	{s,3,2}{6}{4,5}{7}{8}{t}
4	4-6	9	No	ADD	{s,3,2}{4,5,6}{7}{8}{t}
5	2-4	10	No	ADD	{s,3,2,4,5,6}{7}{8}{t}
6	s-7	11	No	ADD	{s,3,2,4,5,6,7}{8}{t}
7	2-6	14	YES	SKIP	2 and 6 in same component
8	3-5	15	YES	SKIP	3 and 5 in same component
9	4-t	16	No	ADD	{s,3,2,4,5,6,7,t}{8}
10	6-t	20	YES	SKIP	6 and t already connected
11	7-8	25	No	ADD	{s,3,2,4,5,6,7,t,8} DONE!
12	8-t	30	YES	SKIP	All vertices connected

Kruskal's MST Total Weight = 3+5+8+9+10+11+16+25 = 67

Reverse Delete Algorithm — remove heaviest edges that don't disconnect graph

Start with ALL edges. Consider descending weight order. Remove if graph stays connected.



Reverse Delete — Decision Table:

Stp	Edge	Wt	Decision	Reason
1	8-t	30	REMOVE	Graph still connected — 8 reachable via 7-8 → s-7
2	7-8	25	KEEP	Would isolate vertex 8 (no other edges to 8 remain)
3	6-t	20	REMOVE	t still connected via 4-t; 6 still via 4-6
4	4-t	16	KEEP	Would disconnect t (8-t already removed)
5	3-5	15	REMOVE	5 reachable via 4-5 → 2-4 → 3-2 → s-3
6	2-6	14	REMOVE	6 still reachable via 4-6
7	s-7	11	KEEP	Would isolate vertex 7 (and 8)
8	2-4	10	KEEP	Would disconnect {s,3,2} from {4,5,6,7,t,8}
9	4-6	9	KEEP	6 isolated — 2-6 already removed
10	3-2	8	KEEP	Would disconnect vertex 2 from s
11	s-3	5	KEEP	Would isolate s from graph
12	4-5	3	KEEP	Would isolate vertex 5

Reverse Delete MST Total Weight = 25+16+11+10+9+8+5+3 = 87

All 3 Algorithms → Same MST = Weight 87

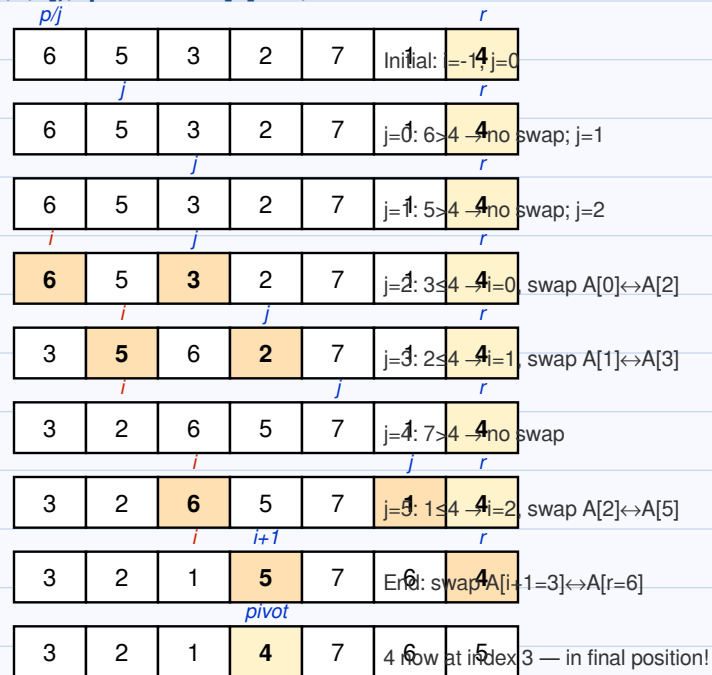
Edges: s-3(5), 3-2(8), 2-4(10), 4-5(3), 4-6(9), s-7(11), 4-t(16), 7-8(25)

(20 pts) 3. Sort $S = \{6, 5, 3, 2, 7, 1, 4\}$ using Quicksort. Show pointer trace.

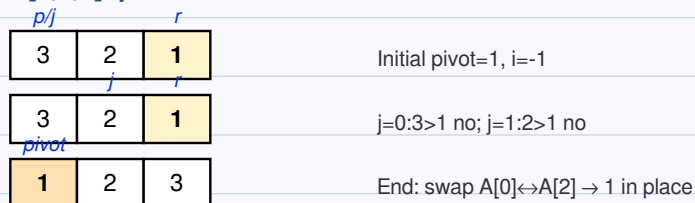
Pivot = last element $A[r]$. i starts at $p-1$. j scans from p to $r-1$.

If $A[j] \leq \text{pivot}$: $i++$, swap $A[i]$ and $A[j]$. After loop: swap $A[i+1]$ and $A[r]$.

PASS 1: PARTITION([6,5,3,2,7,1,4]), pivot $x = A[6] = 4$, $i = -1$

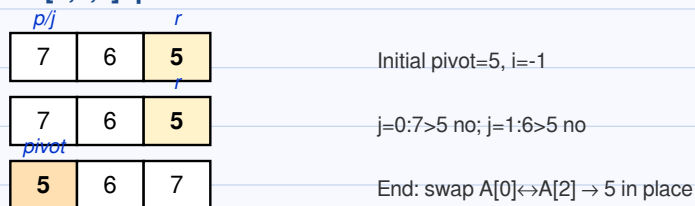


Recurse LEFT [3,2,1] pivot=1:



[2,3]: pivot=3, $2 \leq 3 \rightarrow$ swap $A[0] \leftrightarrow A[0]$, place 3 $\rightarrow$ [2,3] done

Recurse RIGHT [7,6,5] pivot=5:



[6,7]: pivot=7, $6 \leq 7 \rightarrow$ swap $A[0] \leftrightarrow A[0]$, place 7 $\rightarrow$ [6,7] done

1	2	3	4	5	6	7
---	---	---	---	---	---	---

Final Sorted Array: [1, 2, 3, 4, 5, 6, 7]